Reg. No. | | | | | | | | | | | [12EE7610]

SEVENTH SEMESTER B.E EXAMINATION, DECEMBER–2015

EMERGING TECHNOLOGIES IN POWER GENRATION

Time:3 Hours **Maximum Marks:70**

Instructions: Answer any **FIVE** full Questions and all questions carry equal marks.

1.	**a.**	Explain fluidised bed combustion technology for generating clean energy.	**04**
	b.	With a neat figure, explain pressurised FBC.	**06**
	c.	List the advantages of FBC boilers.	**04**
2.	**a.**	What is a fuel cell? Explain its principle and operation.	**05**
	b.	Give the differences between fuel cell and storage battery.	**04**
	c.	What are the advantages of fuel cells?	**05**
3.	**a.**	With a neat sketch, explain polymer electrolyte membrane fuel cell.	**08**
	b.	List out the applications of fuel cell.	**06**
4.	**a.**	List out the common fuels used in fuel cells.	**04**
	b.	Derive an expression for max efficiency of fuel cell.	**04**
	c.	Explain VI characteristic of fuel cell.	**06**
5.	**a.**	Describe the process of electrolytic production of hydrogen.	**08**
	b.	Write short note on:	**06**
		i. Bio photolysis	
		ii. Photo electrolysis	
6.	**a.**	What are the different methods of hydrogen storage? Describe any one method.	**08**
	b.	Explain the causes of fires in hydrogen production.	**06**
7.	**a.**	What is small hydel development?	**06**
	b.	Describe the Bulb turbine used for small scale hydroelectric plants.	**08**
8.	**a.**	What are the main advantages of MHD systems?	**08**
	b.	Discuss the principle of MHD power generation.	**06**

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